

Practice Question Set

1. What is a keyword?
 2. Why are keywords called reserved words?
 3. What is an identifier?
 4. What is a variable?
 5. Why is a variable called a variable?
 6. Write the syntax for variable declaration.
 7. What is variable initialization?
 8. State the five rules for naming identifiers.
 9. What is case sensitivity in C?
 10. What are the types of variables in C?
 11. What is a local variable?
 12. What is a global variable?
 13. What is a static variable?
 14. Differentiate between keywords and identifiers.
 15. Differentiate between declaration and initialization.
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Important Interview/Viva Questions

16. Can keywords be used as variable names? Why?
17. Why can't an identifier start with a number?
18. Is C case-sensitive? Explain with examples.

19. What happens if spaces or special symbols are used in identifiers?

20. Difference between local and global variables.

21. Difference between global and static variables.

22. Explain variable declaration and initialization with examples.

23. Are variables and identifiers the same? Explain.

24. What are the 32 keywords in ANSI C?

25. Which type of variable retains its value between function calls?

Answers

1. What is a keyword?

Answer:

Keywords are reserved words in C having predefined meanings and cannot be used as identifiers.

Examples:

- int
- float
- char
- if
- else
- while

2. Why are keywords called reserved words?

Answer:

Because they are reserved by the compiler for specific purposes and cannot be used as variable or function names.

3. What is an identifier?

Answer:

An identifier is a name given by the programmer to variables, functions, arrays, structures, etc.

Examples:

```
age  
marks  
calculate_salary
```

4. What is a variable?

Answer:

A variable is the name of a memory location used to store data whose value can change during program execution.

5. Why is a variable called a variable?

Answer:

Because its value can vary or change during program execution.

Example:

```
int age = 20;  
age = 21;  
age = 22;
```

6. Write the syntax for variable declaration.

Answer:

```
datatype variable_name;
```

Example:

```
int age;  
float salary;
```

7. What is variable initialization?

Answer:

Assigning a value to a variable at the time of declaration is called initialization.

Example:

```
int age = 20;
```

8. State the five rules for naming identifiers.

Answer:

1. Must start with a letter or underscore.
 2. Cannot start with a digit.
 3. Spaces are not allowed.
 4. Special characters are not allowed.
 5. Keywords cannot be used as identifiers.
 6. C is case-sensitive.
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9. What is case sensitivity in C?

Answer:

Uppercase and lowercase letters are treated differently.

Example:

age

Age

AGE

These are three different identifiers.

10. What are the types of variables in C?

Answer:

1. Local Variable
 2. Global Variable
 3. Static Variable
-

11. What is a local variable?

Answer:

A variable declared inside a function or block is called a local variable.

Example:

```
void fun()
{
    int x;
}
```

12. What is a global variable?

Answer:

A variable declared outside all functions is called a global variable.

Example:

```
int x;

int main()
{
}
```

13. What is a static variable?

Answer:

A variable declared using the static keyword is called a static variable.

Example:

```
static int count;
```

14. Differentiate between Keywords and Identifiers.

Keywords	Identifiers
Reserved words	User-defined names
Predefined meaning	Programmer-defined meaning
Cannot be used as variable names	Used to name variables, functions, arrays etc.
Example: int, float	Example: age, marks

15. Differentiate between Declaration and Initialization.

Declaration	Initialization
Creates a variable	Assigns a value to variable
Memory is allocated	Value is stored
Example: <code>int age;</code>	Example: <code>int age = 20;</code>

Important Interview Answers

16. Can keywords be used as variable names?

Answer:

No. Keywords are reserved words and cannot be used as identifiers.

✗ Invalid:

```
int int;  
float if;
```

17. Why can't an identifier start with a number?

Answer:

Because the compiler may confuse it with a numeric constant.

✗ Invalid:

```
1age  
2salary
```

✓ Valid:

```
age1  
salary2
```

18. Is C case-sensitive?

Answer:

Yes. C treats uppercase and lowercase letters differently.

Example:

age
Age
AGE

All are different variables.

19. What happens if spaces or special symbols are used in identifiers?

Answer:

The compiler generates a syntax error.

✗ Invalid:

student age
marks%
salary@

✓ Valid:

student_age
studentAge

20. Difference between Local and Global Variables

Local Variable	Global Variable
Declared inside function	Declared outside function
Accessible only inside that function	Accessible throughout the program
Lifetime limited to function execution	Lifetime is entire program

21. Difference between Global and Static Variables

Global Variable	Static Variable
Accessible throughout program	Scope depends on declaration
Lifetime entire program	Lifetime entire program
Declared normally	Declared using static keyword

22. Explain declaration and initialization with examples.

Answer:

Declaration:

```
int age;
```

Initialization:

```
int age = 20;
```

23. Are variables and identifiers the same?

Answer:

No.

- An identifier is a name given to variables, functions, arrays, structures, etc.
- A variable is a specific type of identifier used to store data.

Thus, **every variable is an identifier, but every identifier is not necessarily a variable.**

24. How many keywords are there in ANSI C?

Answer:

ANSI C contains **32 keywords**.

25. Which type of variable retains its value between function calls?

Answer:

A **static variable** retains its value between function calls.

Example:

```
static int count = 0;
```